Supplementary to "SpatialMuSC: Spatial Graph-Conditioned Diffusion Refinement for Robust Spatial Multi-Omics Integration"

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1. Implementation details of SpatialMuSC

1.1 Graph construction

For each modality m, SpatialMuSC constructs two neighborhood graphs: a spatial graph and a feature graph. The spatial graph is constructed from the spot coordinates C using a k_s-nearest-neighbor strategy. Specifically, for each spot i, its spatial neighbor set $\mathcal{N}_{s,m}(i)$ is determined according to Euclidean distance in the coordinate space. When all modalities share the same spatial coordinates, the spatial neighbor sets are identical across modalities, but the modality index m is retained for notational consistency.

The feature graph is constructed independently for each modality using the molecular profile X_m. Before graph construction, modality-specific features are preprocessed and normalized according to the data type. The feature neighbor set $\mathcal{N}_{f,m}(i)$ is obtained by searching the k_f nearest spots in the feature space using cosine similarity or Euclidean distance. The resulting spatial and feature graphs are symmetrized, augmented with self-loops and normalized as described in the main text.

1.2 Local topology-preserving encoder

The local topology-preserving module contains two graph-view encoders for each modality: one operating on the spatial graph and the other on the feature graph. Each graph-view encoder consists of graph propagation, linear projection and nonlinear activation. The two graph-view embeddings are concatenated and passed through a projection network to obtain the local topology-preserving representation L_m.

In our implementation, the projection network $\psi_m(\cdot)$ is implemented as a multilayer perceptron:

$$\psi_m(Q) = \sigma(QW^{\psi}_{m,1} + b^{\psi}_{m,1})W^{\psi}_{m,2} + b^{\psi}_{m,2}, \tag{1}$$

where Q denotes the concatenated spatial- and feature-graph embeddings, $W^{\psi}_{m,1}$ and $W^{\psi}_{m,2}$ are learnable weight matrices, $b^{\psi}_{m,1}$ and $b^{\psi}_{m,2}$ are bias terms, and $\sigma(\cdot)$ denotes a nonlinear activation function. This projection compresses complementary neighborhood information into a unified local tissue representation.

1.3 Global encoder architecture

The global encoder $\eta_m(\cdot)$ maps the modality-specific feature matrix X_m into the initial global latent representation G^0_m. In our implementation, $\eta_m(\cdot)$ first projects X_m into a shared hidden space and then models spot-wise global dependencies using self-attention layers.

Let S^0_m denote the projected input sequence:

$$S^0_m = X_m P_m, \tag{2}$$

where $P_m \in \mathbb{R}^{d_m \times c}$ is a learnable projection matrix and c is the hidden dimension.

For the ℓ-th self-attention layer, the query, key and value matrices are computed as

$$Q^{\ell}_m = S^{\ell-1}_m W^{\ell}_{Q,m}, \qquad K^{\ell}_m = S^{\ell-1}_m W^{\ell}_{K,m}, \qquad V^{\ell}_m = S^{\ell-1}_m W^{\ell}_{V,m}, \tag{3}$$

where $W^{\ell}_{Q,m}$, $W^{\ell}_{K,m}$ and $W^{\ell}_{V,m}$ are learnable projection matrices.

The self-attention output is obtained by

$$\mathrm{Attn}^{\ell}_m = \mathrm{softmax}\left(\frac{Q^{\ell}_m (K^{\ell}_m)^T}{\sqrt{d_a}}\right) V^{\ell}_m, \tag{4}$$

where d_a is the hidden dimension of each attention head. Residual connections, feed-forward transformations and normalization layers are applied after the attention operation. The final hidden representation is projected to obtain the initial global latent representation G^0_m.

1.4 Graph-aware denoising implementation

The main text defines the spatial graph-conditioned diffusion refinement process. Here, we describe the implementation of the graph-aware denoising network. At diffusion step t, the noisy latent representation G_m^t is first propagated over the normalized spatial graph to obtain the graph-conditioned noisy latent state:

$$\tilde{G}_m^t = \tilde{A}_m^s G_m^t. \tag{5}$$

The diffusion step t is encoded by a learnable time embedding function $e(t)$ and broadcast to all spots. The denoising network first combines the graph-conditioned noisy latent state with the diffusion time embedding:

$$Q_m^t = \sigma\left(\tilde{G}_m^t W_{m,1}^d + \mathbf{1}e(t)^T W_{m,t}^d + b_{m,1}^d\right), \tag{6}$$

where $\mathbf{1} \in \mathbb{R}^N$ is an all-one vector used to broadcast the time embedding to all spots.

The hidden state is then further propagated over the spatial graph for topology-aware noise estimation:

$$\bar{Q}_m^t = \tilde{A}_m^s Q_m^t. \tag{7}$$

The predicted noise is obtained by

$$\hat{\epsilon}_m^t = \mathrm{LN}\left(\bar{Q}_m^t W_{m,2}^d + b_{m,2}^d\right), \tag{8}$$

where $W_{m,1}^d$, $W_{m,t}^d$ and $W_{m,2}^d$ are learnable weight matrices, $b_{m,1}^d$ and $b_{m,2}^d$ are bias terms, $\sigma(\cdot)$ denotes a nonlinear activation function, and $\mathrm{LN}(\cdot)$ denotes layer normalization. This design enables spatial topology to affect noise prediction both through the graph-conditioned noisy latent state and through graph-aware propagation within the denoising network.

1.5 Deterministic reverse refinement

During training, SpatialMuSC optimizes the DDPM-style noise prediction objective described in the main text. During inference, SpatialMuSC does not generate latent representations from pure Gaussian noise. Instead, it starts from the encoder-derived latent representation and applies deterministic reverse refinement to obtain a denoised latent representation.

Given the predicted noise $\hat{\epsilon}_m^t$, a DDPM-style deterministic reverse update can be written as

$$G_m^{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(G_m^t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\epsilon}_m^t\right). \tag{9}$$

The stochastic sampling term is omitted to obtain stable latent representations.

Equivalently, the denoised latent representation at step t can be estimated as

$$\hat{G}_m^{0,t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(G_m^t - \sqrt{1-\bar{\alpha}_t}\hat{\epsilon}_m^t\right). \tag{10}$$

The final refined representation is defined as

$$G_m = \hat{G}_m^{0,t}. \tag{11}$$

In all experiments, we used one deterministic reverse refinement step during inference. This setting is intended to refine encoder-derived latent representations for stable downstream clustering, rather than to perform stochastic generation from pure noise.

1.6 Cross-scale alignment implementation

Cross-scale alignment couples the local topology-preserving representation L_m and the refined global representation G_m. The projection heads $\rho_L(\cdot)$ and $\rho_G(\cdot)$ are implemented as two-layer multilayer perceptrons:

$$\rho_L(L_m) = \sigma(L_m W_L^{(1)} + b_L^{(1)})W_L^{(2)} + b_L^{(2)}, \tag{12}$$

$$\rho_G(G_m) = \sigma(G_m W_G^{(1)} + b_G^{(1)})W_G^{(2)} + b_G^{(2)}. \tag{13}$$

The spatially weighted positive matrix P_m is constructed according to the spatial graph. The same spot is assigned the strongest positive weight, and its spatial neighbors are assigned soft-positive weights. Other non-neighboring spots are treated as negative samples. This design encourages local and global representations to be aligned not only at the same spot but also within spatially coherent neighborhoods.

1.7 Multi-omic fusion network

For each modality, the local representation L_m and the refined global representation G_m are concatenated and projected into a modality-specific embedding H_m. The projection network $\phi_m(\cdot)$ is implemented as

$$H_m = \sigma([L_m \parallel G_m]W^h_{m,1} + b^h_{m,1})W^h_{m,2} + b^h_{m,2}, \tag{14}$$

where $W^h_{m,1}$ and $W^h_{m,2}$ are learnable weight matrices, and $b^h_{m,1}$ and $b^h_{m,2}$ are bias terms.

The modality-specific embeddings are then concatenated and passed through the multi-omic fusion network:

$$Z = \sigma([H_1 \parallel H_2 \parallel \cdots \parallel H_M]W^z_1 + b^z_1)W^z_2 + b^z_2. \tag{15}$$

The fused representation Z is used for clustering-based spatial domain identification and downstream biological interpretation.

1.8 Perturbation-invariant fusion consistency implementation

Perturbation-invariant fusion consistency is used to regularize the multi-omic fusion space. It contains two complementary components: cross-modal consistency and fusion-representation consistency. Cross-modal consistency encourages modality-specific embeddings H_m and H_n to agree on shared tissue organization, whereas fusion-representation consistency constrains the fused representation Z to remain consistent under perturbed observations.

For fusion-representation consistency, a stochastic perturbation operator $C(\cdot)$ is applied to the input features to generate perturbed observations:

$$X^+_m = C(X_m). \tag{16}$$

In our implementation, $C(\cdot)$ is implemented as feature-level stochastic perturbation:

$$X^+_m = X_m \odot B_m, \tag{17}$$

where $B_m \in \{0,1\}^{d_m}$ is sampled from a Bernoulli distribution with feature keep ratio r_m, and $\odot$ denotes element-wise multiplication. The same mask is shared across spots within modality m, mimicking feature-level corruption or partial missing observations.

The perturbed observations $\{X^+_m\}^M_{m=1}$ are fed into the same SpatialMuSC model to obtain a perturbed fused representation Z^+. The original fused representation Z and perturbed fused representation Z^+ are then aligned using the symmetric contrastive objective defined in the main text. The final perturbation-invariant fusion consistency loss is denoted as $\mathcal{L}_{\mathrm{pif}}$.

1.9 Reconstruction decoder

The modality-specific reconstruction decoder maps the fused representation back to the original feature space of each modality. In our implementation, the decoder consists of a linear projection followed by spatial graph propagation:

$$\widehat{X}_m = \tilde{A}^s_m Z W^r_m, \tag{18}$$

where $\tilde{A}^s_m$ is the normalized spatial adjacency matrix and W^r_m is a learnable decoder weight matrix. The reconstruction objective encourages the fused representation to retain molecular information from each omics layer.

1.10 Reproducibility settings and ablation definitions

To improve reproducibility, we summarize the implementation settings used in SpatialMuSC. In our implementation, SpatialMuSC was trained using the Adam optimizer with a learning rate of 1×10^{-4} and weight decay of 0. The output embedding dimension was set to 64. Unless otherwise specified, the random seed was set to 2022.

For graph construction, the spatial and feature graphs were generated by the preprocessing function `adjacent_matrix_preprocessing`. The spatial graph was constructed from spot coordinates, and the feature graph was constructed from the preprocessed molecular features of each modality. Both spatial and feature adjacency matrices were used in the local topology-preserving branch. The neighbor numbers k_s and k_f were set according to the graph construction settings in the preprocessing module.

The global encoder was implemented as a Transformer encoder over spots, where spots were treated as tokens. The default Transformer hidden dimension was 128, with 2 encoder layers and 8 attention heads. For the Stereo-CITE-seq dataset, the number of attention heads was set to 4. Each Transformer block contained multi-head self-attention, residual connections, layer normalization and a feed-forward network with GELU activation. The Transformer output was projected to a 64-dimensional latent representation for each modality.

For diffusion refinement, a DDPM-style forward noising and noise-prediction objective was applied to the Transformer-derived global representations. During training, the diffusion step t was sampled uniformly from $\{1, \ldots, T\}$ for each spot. We used $T = 500$ diffusion steps and a linear beta schedule from $\beta_1 = 1 \times 10^{-4}$ to $\beta_T = 2 \times 10^{-2}$. Gaussian noise was injected according to

$$G_m^t = \sqrt{\bar{\alpha}_t} G_m^0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{19}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{\tau=1}^{t} \alpha_\tau$.

The graph-aware denoising network used a 128-dimensional sinusoidal time embedding. The noisy latent representation was first propagated over the normalized spatial graph to obtain the graph-conditioned noisy latent state. The graph-conditioned noisy latent state was combined with the diffusion time embedding, further propagated over the spatial graph, and then passed through linear projection and layer normalization to predict the injected noise. The diffusion loss was computed as the mean squared error between the injected noise and the predicted noise. During inference, SpatialMuSC did not generate latent representations from pure Gaussian noise. Instead, it started from the encoder-derived global latent representation and applied deterministic reverse refinement. The stochastic sampling term was omitted to obtain stable latent representations for clustering. The number of reverse refinement steps was set to 1 in all experiments.

For perturbation-invariant fusion consistency, the perturbation operator was implemented as feature-level random masking. Specifically, for each modality, a Bernoulli feature mask was sampled and shared across all spots within that modality:

$$X_m^+ = X_m \odot B_m, \tag{20}$$

where $B_m \in \{0, 1\}^{d_m}$ denotes the feature mask. The feature keep ratio was set to $r = 0.2$ in the main experiments. Thus, a proportion r of features was retained, and the remaining features were masked. The perturbed observations were fed into the same SpatialMuSC model to obtain the perturbed fused representation Z^+.

The fusion-level contrastive consistency loss was computed using full-batch in-batch negatives. Given the fused representation Z and the perturbed fused representation Z^+, both representations were first normalized, and an $N \times N$ similarity matrix was computed between all spots. The diagonal entries were treated as positive pairs, and all other spots in the batch were used as negatives. The loss was computed symmetrically between Z and Z^+.

We summarize the definitions of the ablated and targeted variants used in the ablation study. The w/o Diff variant removes the entire diffusion refinement module. The w/o CrossScale variant removes cross-scale alignment between local topology-preserving representations and refined global representations. The w/o PIF variant removes perturbation-invariant fusion consistency. The w/o GraphCond variant retains diffusion refinement but removes spatial graph conditioning from the denoising process by replacing the graph-conditioned noisy latent state with the original noisy latent state:

$$\tilde{G}_m^t = G_m^t. \tag{21}$$

The Diff+PostSmooth variant performs ordinary diffusion refinement without spatial graph conditioning during noise prediction and then applies post hoc graph smoothing to the refined latent representation:

$$G_{m,\text{post}} = \tilde{A}_m^s G_{m,\text{diff}}, \tag{22}$$

where $G_{m,\text{diff}}$ denotes the representation obtained from ordinary diffusion refinement. The GraphCond-only variant retains spatial graph propagation but removes Gaussian noise injection, diffusion noise prediction, diffusion loss and reverse refinement. This variant tests whether graph propagation alone is sufficient to explain the performance gain of SpatialMuSC. Together, these targeted variants distinguish spatial graph-conditioned diffusion refinement from ordinary diffusion refinement, post hoc graph smoothing and graph propagation alone.

2. Data

2.1 Human lymph node dataset

This 10x Genomics-based spatial transcriptome–proteome dataset comprises 3,484 spots, with 18,085 genes measured in the RNA modality and 31 proteins measured in the ADT modality. Manual annotations provided by SpatialGlue [1] were used as reference labels. The anatomical domains are spatially discrete, making this dataset suitable for evaluating spatial domain identification, boundary delineation and cross-modality consistency in a complex human lymphoid microenvironment.

2.2 Mouse thymus dataset

This Stereo-CITE-seq spatial transcriptome–proteome dataset comprises 4,697 spots, with 23,622 genes measured in the RNA modality and 51 proteins measured in the ADT modality. The highly organized thymic architecture and distinct immune niches make this dataset suitable for evaluating whether integrative methods can preserve fine-grained spatial organization while capturing complementary transcriptomic and proteomic information.

2.3 Mouse spleen dataset

This SPOTS spatial transcriptome–proteome dataset comprises 2,568 spots, with 32,285 genes measured in the RNA modality and 21 proteins measured in the protein modality. The structured immune compartments of the spleen make this dataset suitable for evaluating spatial domain identification and multimodal representation learning in lymphoid tissue.

2.4 Adult human brain hippocampus dataset

This spatial-epigenome-transcriptome dataset comprises 2,500 spatial pixels, with 21,179 genes measured in the RNA modality and 56,614 chromatin-accessibility features measured in the ATAC modality. Manual annotations provided by MultiGATE [2] were used as reference labels. In contrast to the human lymph node dataset, the annotated domains are relatively continuous, making this dataset suitable for evaluating whether integrative methods can capture gradual spatial transitions and modality-consistent molecular variation in a heterogeneous neural microenvironment.

3. Evaluation metrics

To quantitatively evaluate the agreement between the predicted clustering and the reference annotation, we used eight complementary external clustering metrics, including adjusted Rand index (ARI) [3], normalized mutual information (NMI) [4], adjusted mutual information (AMI) [5], Fowlkes–Mallows index (FMI) [6], V-measure [7], F1-score [8], Jaccard [9] and completeness [7].

Let $Y = \{y_i\}_{i=1}^{N}$ and $\hat{Y} = \{\hat{y}_i\}_{i=1}^{N}$ denote the ground-truth and predicted labels for N spots, respectively, and let n_{ij} denote the number of spots simultaneously assigned to ground-truth class i and predicted cluster j. We first assessed pairwise clustering agreement using the adjusted Rand index, which corrects the Rand index for chance agreement:

$$\mathrm{ARI} = \frac{\sum_{i,j}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] \Big/ \binom{N}{2}}{\left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] \Big/ 2 - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] \Big/ \binom{N}{2}} \tag{23}$$

where $a_i = \sum_j n_{ij}$ and $b_j = \sum_i n_{ij}$ denote the marginal counts of the contingency table. We next quantified the shared information between the two partitions using normalized mutual information:

$$\mathrm{NMI} = \frac{2I(Y;\hat{Y})}{H(Y) + H(\hat{Y})} \tag{24}$$

where $I(Y;\hat{Y})$ denotes the mutual information between Y and $\hat{Y}$, and $H(\cdot)$ denotes entropy. To further account for agreement expected by chance, we computed the adjusted mutual information:

$$\mathrm{AMI} = \frac{I(Y;\hat{Y}) - E[I(Y;\hat{Y})]}{\left[H(Y) + H(\hat{Y})\right]/2 - E[I(Y;\hat{Y})]} \tag{25}$$

where $E[I(Y;\hat{Y})]$ is the expected mutual information under random labeling with fixed cluster-size marginals. We also measured the geometric mean of pairwise precision and recall using the Fowlkes–Mallows index:

$$\mathrm{FMI} = \frac{TP}{\sqrt{(TP + FP)(TP + FN)}} \tag{26}$$

where TP, FP and FN denote the numbers of true-positive, false-positive and false-negative spot pairs, respectively. To summarize clustering quality from a conditional-entropy perspective, we used the V-measure:

$$\text{V-measure} = \frac{2hc}{h + c} \tag{27}$$

where h and c denote homogeneity and completeness, respectively. In parallel, we evaluated pairwise co-clustering consistency using the F1-score:

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{28}$$

where the F1-score here is the pairwise F1-score computed from co-assigned spot pairs. We further quantified the overlap between correctly identified co-clustered pairs and all discordant pair assignments using the Jaccard index:

$$\mathrm{Jaccard} = \frac{TP}{TP + FP + FN} \tag{29}$$

Finally, we reported completeness to measure whether spots from the same reference class were assigned to a single predicted cluster:

$$\mathrm{Completeness} = 1 - \frac{H(\hat{Y} \mid Y)}{H(\hat{Y})} \tag{30}$$

where $H(\hat{Y} \mid Y)$ denotes the conditional entropy of the predicted partition given the reference annotation. For all eight metrics, larger values indicate better agreement between the inferred clustering and the ground-truth labels.

Supplementary Figures

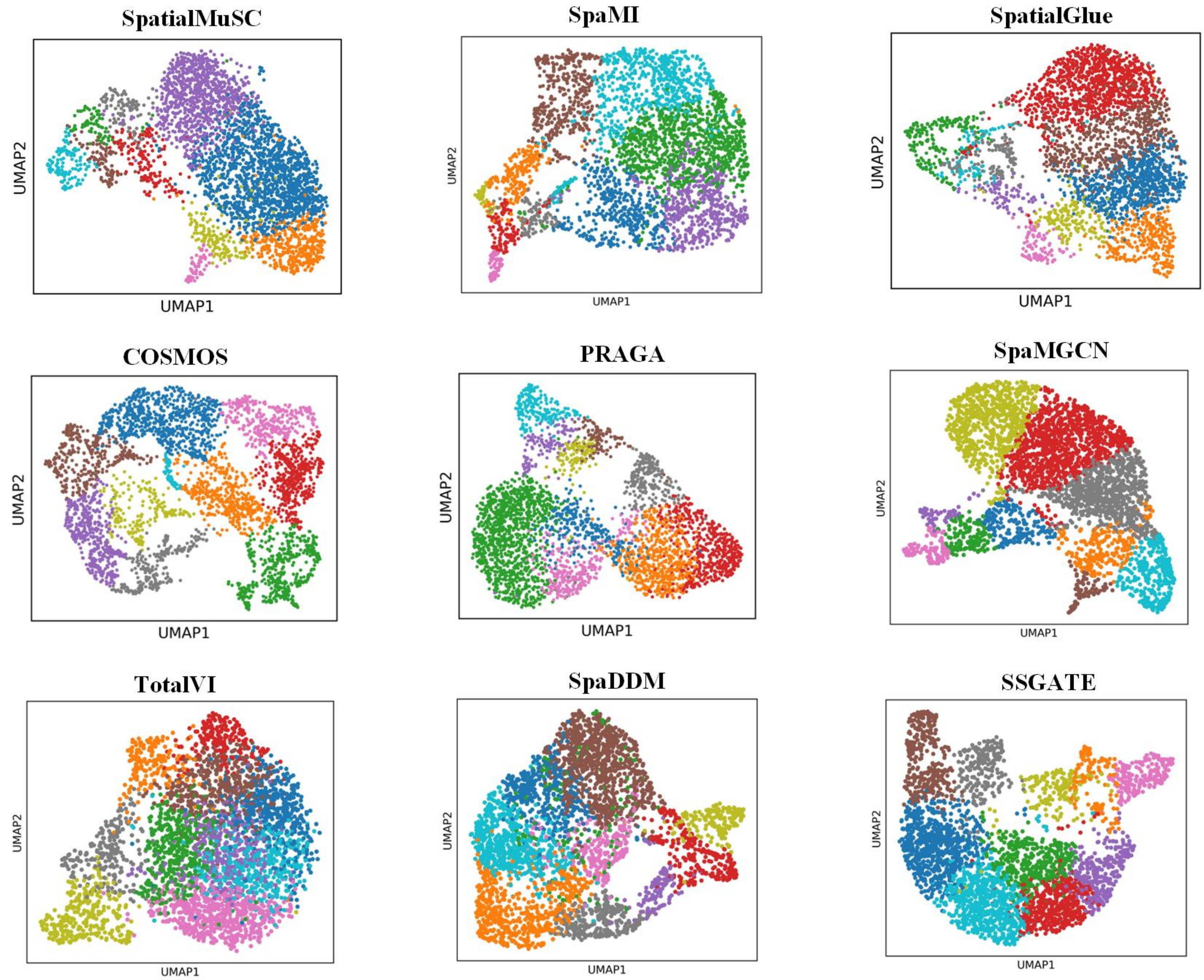

Figure S1: UMAP visualization of spatial domain predictions by SpatialMuSC and other methods on the human lymph node A1 dataset with a discrete spatial organization.

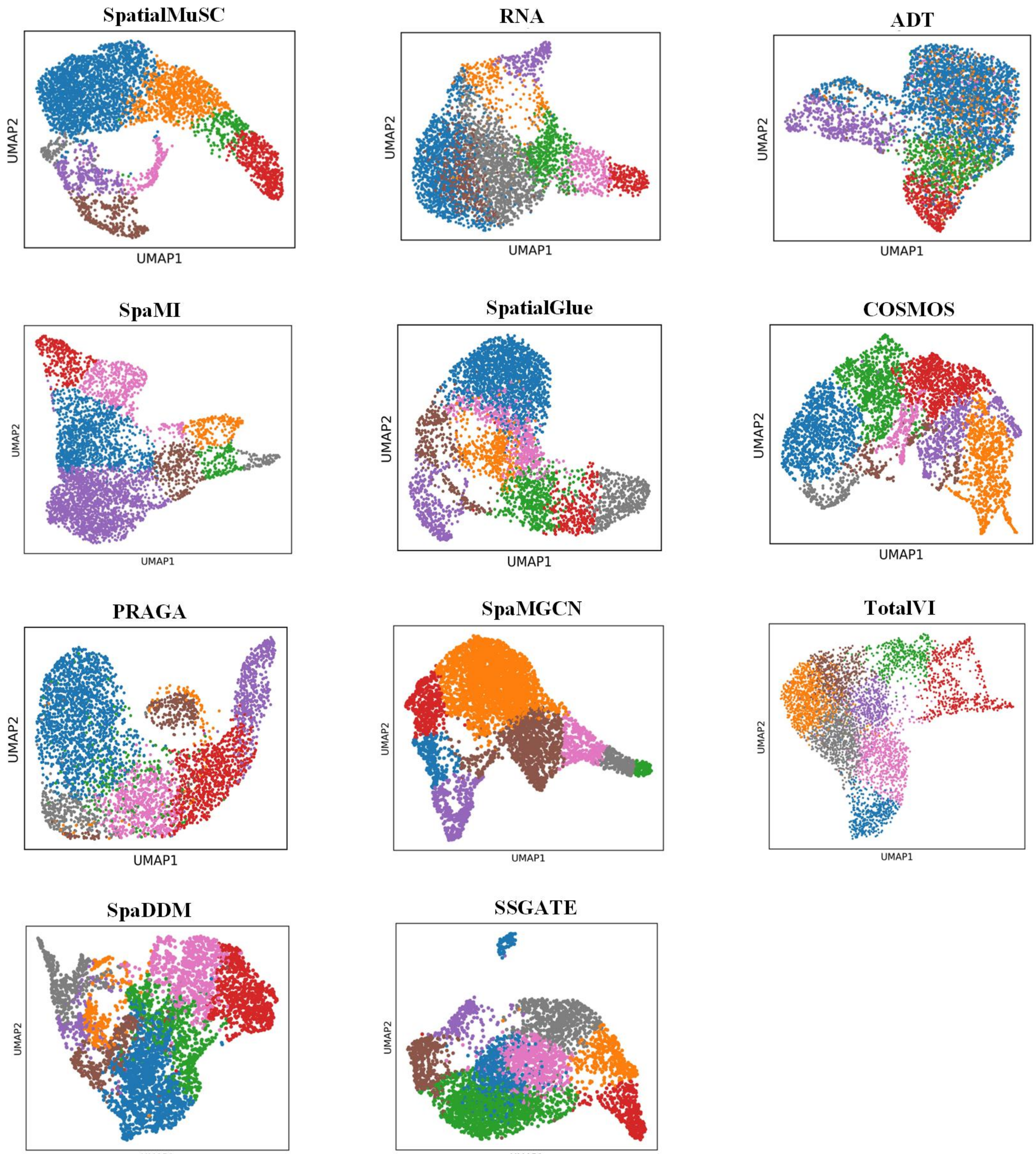

Figure S2: UMAP visualization of spatial domain predictions by SpatialMuSC and other methods on the mouse thymus dataset with a layered spatial organization.

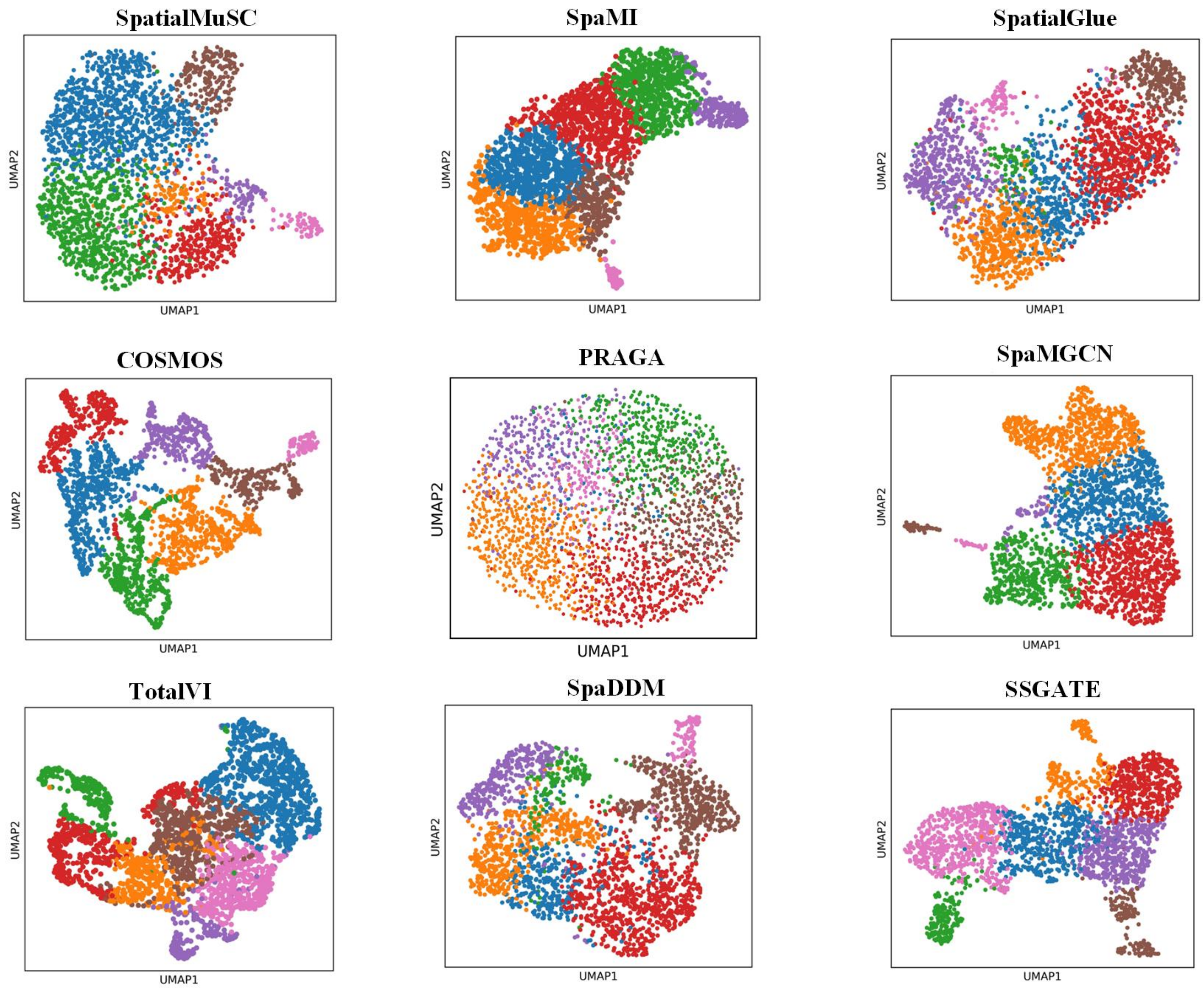

Figure S3: UMAP visualization of spatial domain predictions by SpatialMuSC and other methods on the adult human hippocampus dataset with a continuous spatial organization.

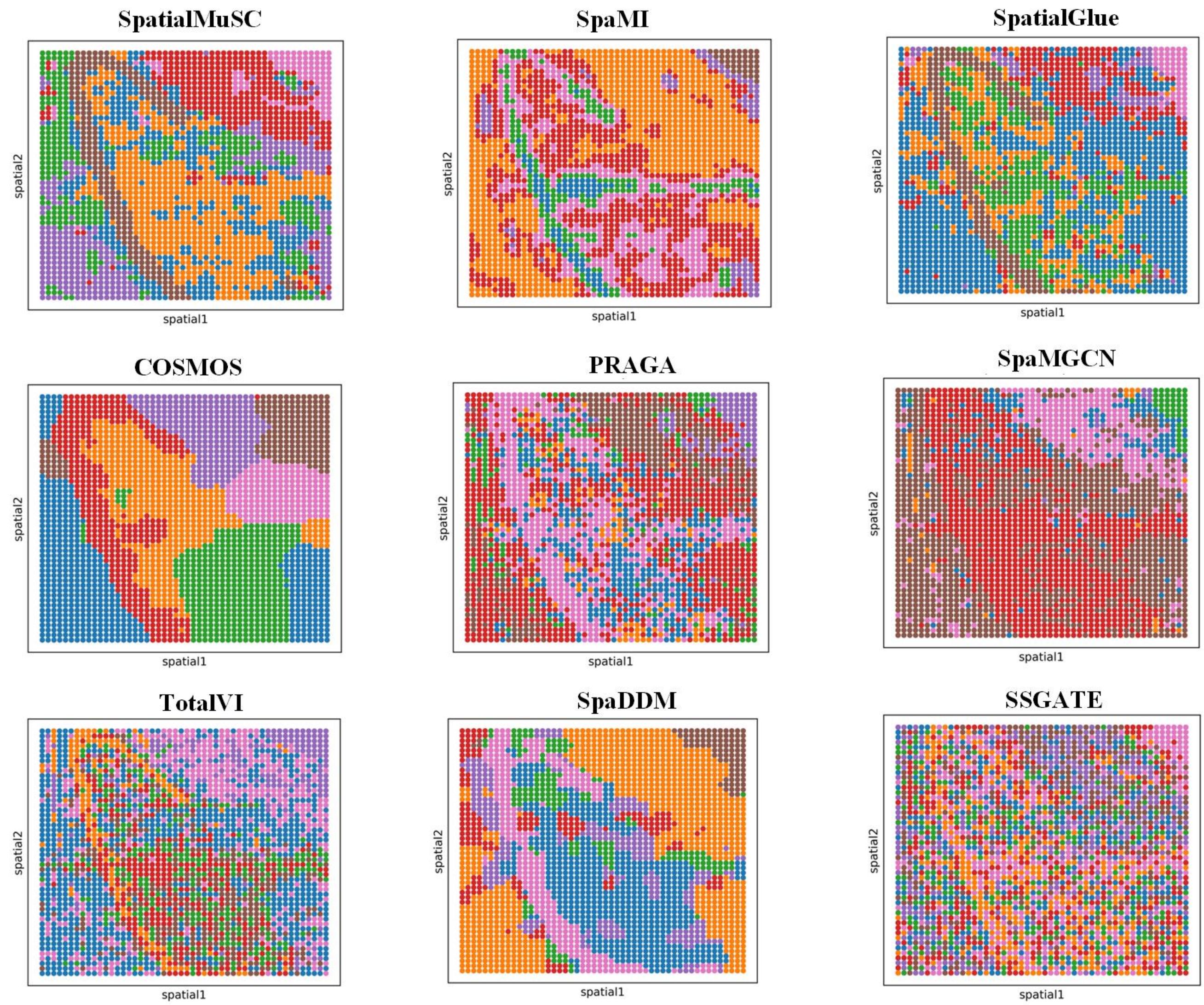

Figure S4: Visualization of spatial domain predictions by SpatialMuSC and other methods on the 50%-masked adult human hippocampus dataset with a continuous spatial organization.

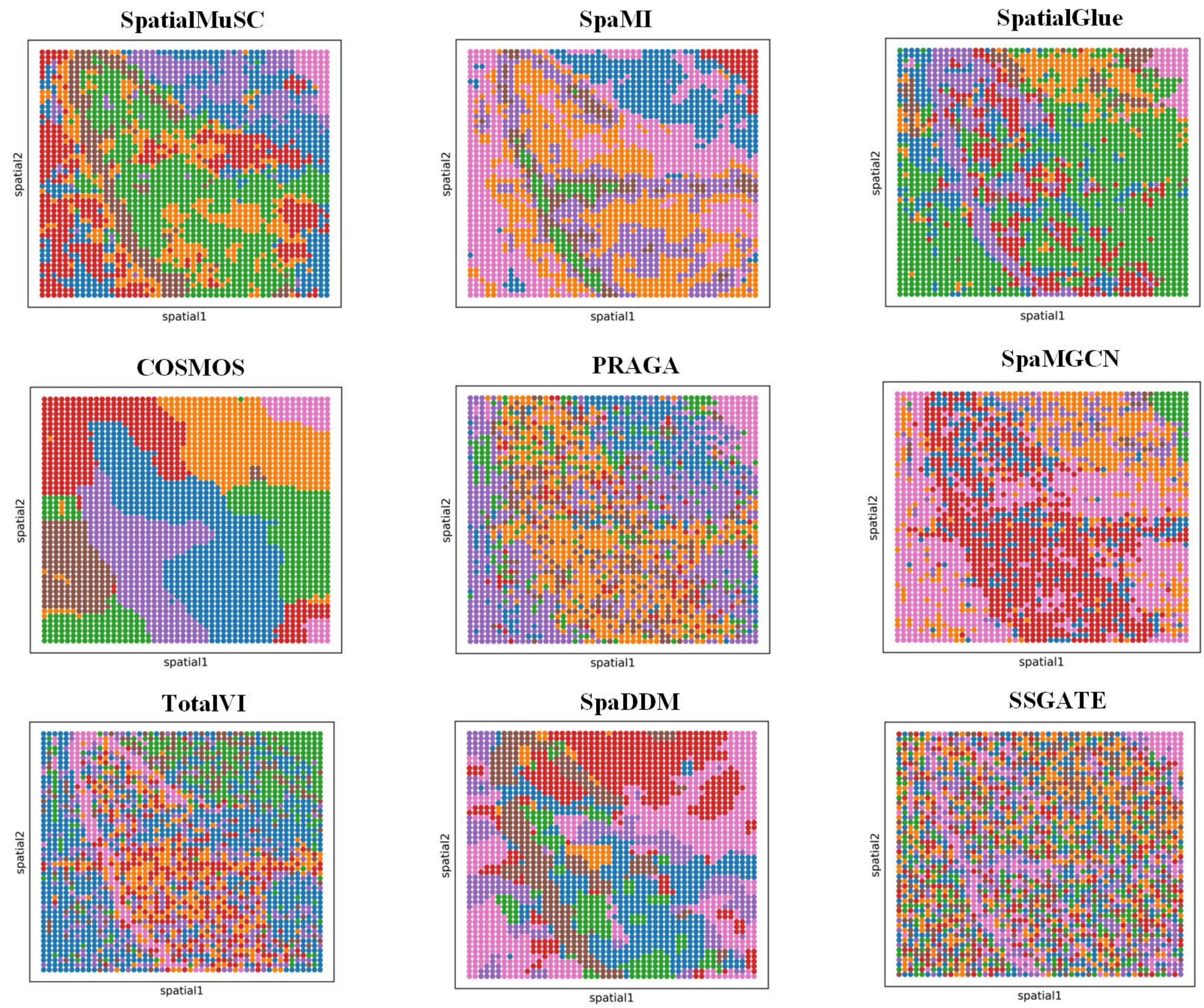

Figure S5: Visualization of spatial domain predictions by SpatialMuSC and other methods on the 70%-masked adult human hippocampus dataset with a continuous spatial organization.

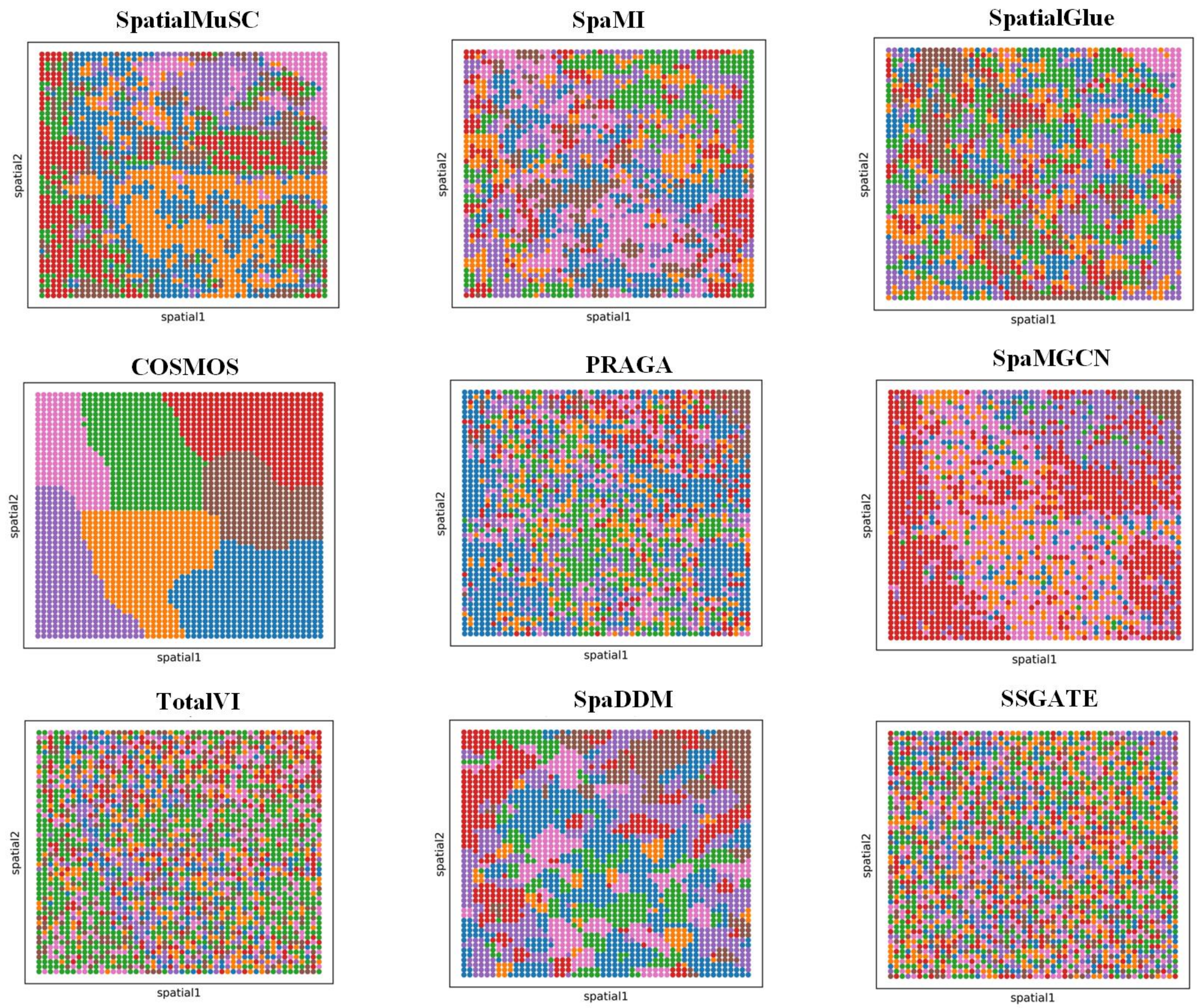

Figure S6: Visualization of spatial domain predictions by SpatialMuSC and other methods on the 90%-masked adult human hippocampus dataset with a continuous spatial organization.

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